eml★ — Minimal Anti-Holomorphic Extension of the EML Sheffer Operator

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GitHub: https://github.com/antparis/eml_star

Abstract

Odrzywołek (2026) showed that $\mathrm{eml}(x, y) = e^x - \ln y$, together with the constant 1, generates all standard elementary functions via finite composition. We identify a structural limitation: eml is holomorphic, so complex conjugation $\bar{z}$, and real and imaginary parts are not reachable by finite eml-compositions. We introduce the companion operator $\mathrm{eml}★(x, y) = e^x - \ln \bar{y}$, which acts as a mirror reflecting the imaginary axis. We prove: (i) $\bar{z} = 1 - \mathrm{eml}★(0, \mathrm{eml}(z, 1))$ at depth 2, conditional on $\mathrm{Im}(z) \in [-\pi, \pi)$; (ii) {eml, eml★, 1} is dense in $C(K, \mathbb{C})$ for every compact $K \subset \{z : \mathrm{Im}(z) \in [-\pi, \pi)\}$ by Stone–Weierstrass; (iii) the exact branch limitation is $\mathrm{Im}(z) \in [-\pi, \pi)$. A direct numerical experiment confirms Theorem 3.1 to machine precision: eml★ achieves MSE = 5.89×10^{-33} vs. 12.97 for eml alone — a ratio of 2.2×10^{33}. Theorems 2.1, 4.3, and Corollary 2.2 are unconditional.

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1. Introduction

Odrzywołek [1] demonstrated that $\mathrm{eml}(x, y) = e^x - \ln y$, together with the constant 1, is a continuous Sheffer operator: every standard elementary function is a finite composition of eml-nodes. The grammar is $S \to 1 \mid \mathrm{eml}(S, S)$. This is the continuous analogue of the NAND gate in Boolean logic.

A natural question arises: does {eml, 1} reach all continuous functions on $\mathbb{C}$, or only the holomorphic ones? The answer is: only the holomorphic ones. This note identifies the obstruction precisely and proposes a minimal fix — the companion operator eml★ — that closes the gap.

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2. The Holomorphic Barrier

Let $E_0 = \{1\}$, $E_{n+1} = E_n \cup \{\mathrm{eml}(f, g) : f, g \in E_n\}$, and $E = \cup_n E_n$.

Theorem 2.1 (Holomorphic closure). *Every element of E is holomorphic on its domain.*

Proof. The constant 1 is holomorphic. The maps $z \mapsto e^z$ and $z \mapsto \ln z$ (principal branch) are holomorphic. Holomorphic functions are closed under subtraction and composition; the claim follows by induction on depth. ∎

Corollary 2.2. *The operators Re, Im, and $z \mapsto \bar{z}$ are not in E.*

Proof. These maps are non-holomorphic (∂_{z̄} Im(z) = 1/2 ≠ 0), so no element of E equals them by Theorem 2.1. ∎

Remark 2.3. Corollary 2.2 does not imply $\pi \notin E$. Odrzywołek shows $\pi \in E$ at depth $K > 53$ (Table 4 of [1]). The distinction is between Im(·) as an operator and π as a fixed real constant.

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3. The Companion Operator eml★

Motivation. eml cannot distinguish z from z̄ — it only sees the holomorphic world. To recover the conjugate, one modification suffices: replace Log(y) by Log(ȳ) in the second argument.

Definition. $\mathrm{eml}{\bigstar}(x, y) := e^x - \mathrm{Log}(\bar{y})$, where Log denotes the principal logarithm ($\mathrm{Arg} \in (-\pi, \pi]$).

Theorem 3.1 (Conjugate recovery at depth 2). *For all $z \in \mathbb{C}$ with $\mathrm{Im}(z) \in [-\pi, \pi)$:*

> $\bar{z} = 1 - \mathrm{eml}{\bigstar}(\ 0, \mathrm{eml}(z, 1)\ )$

Proof. $\mathrm{eml}(z, 1) = e^z$. Then $\mathrm{eml}{\bigstar}(0, e^z) = 1 - \ln(\mathrm{conj}(e^z)) = 1 - \bar{z}$, provided $\mathrm{Im}(\bar{z}) = -\mathrm{Im}(z) \in (-\pi, \pi]$, i.e. $\mathrm{Im}(z) \in [-\pi, \pi)$. ∎

Theorem 3.2 (Exact safe domain). The identity of Theorem 3.1 holds if and only if $\mathrm{Im}(z) \in [-\pi, \pi)$. At $\mathrm{Im}(z) = \pi$ the principal Log produces a jump of $2\pi i$. The boundary $\mathrm{Im}(z) = -\pi$ holds; $\mathrm{Im}(z) = +\pi$ fails. Outside the strip, jumps of exactly $2\pi i$ occur.

Corollary 3.3. $|z|$ is reachable at depth 4 via $|z| = \sqrt{(z \cdot \bar{z})}$, where $\bar{z}$ comes from Theorem 3.1 and multiplication from the eml arithmetic of [1].

Depth table

Expression	Rel. depth*	Via
$\bar{z} = 1 - \mathrm{eml}{\bigstar}(0, \mathrm{eml}(z,1))$	2	Theorem 3.1
$\mathrm{Re}(z) = (z + \bar{z})/2$	3	Corollary 3.3
$\|z\|^2 = z \cdot \bar{z}$	3	eml arithmetic + Thm 3.1
$\|z\| = \sqrt{(z \cdot \bar{z})}$	4	sqrt via eml [1]

Relative depth counts eml★ nodes only.

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4. Topological Completeness via Stone–Weierstrass

Lemma 4.1. *Every rational number lies in E.*

Proof. 0 = eml(0, 1). 1 is given. Arithmetic follows from [1]. ∎

Lemma 4.2. *Every polynomial in $\mathbb{Q}[z, \bar{z}]$ lies in the algebra generated by {eml, eml★, 1}.*

Proof. z ∈ E. $\bar{z}$ by Theorem 3.1. Products and sums by eml arithmetic. ∎

Theorem 4.3 (Stone–Weierstrass density). *For every compact $K \subset \mathbb{C}$ with $K \subset \{z : Im(z) \in [-\pi, \pi)\}$, the algebra generated by {eml, eml★, 1} is dense in $C(K, \mathbb{C})$.*

Proof. On K, Theorem 3.1 gives $\bar{z}$ exactly, so $A = \mathbb{Q}[z, \bar{z}] \subseteq$ {eml, eml★, 1} pointwise. A satisfies Stone–Weierstrass [2]: (i) separates points; (ii) does not vanish; (iii) closed under conjugation. Hence A is dense in $C(K, \mathbb{C})$. ∎

Remark 4.4. Theorem 4.3 does not contradict Theorem 2.1: the algebra is dense, but individual elements of E remain holomorphic. Density is an approximation result, not a representation result.

Remark 4.5 (Global extension). The restriction $Im(z) \in [-\pi, \pi)$ is essential. Density on arbitrary compacts in $\mathbb{C}$ would require tracking winding numbers continuously — impossible with finite compositions of analytic/anti-analytic operators (monodromy theorem). Open problem.

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5. Numerical Experiment

We provide a direct machine-precision verification of Theorem 3.1 and contrast eml★ with eml alone. The test is deterministic, parameter-free, and fully reproducible from the repository.

Setup. N = 40 points per axis, $Re(z) \in [-3, 3]$, $Im(z) \in [-\pi + 0.1, \pi - 0.1]$ (strictly inside the safe strip). Total: 1600 evaluation points. Target: $\bar{z}$.

Test 1 — with eml★ (Theorem 3.1):

```
pred = 1 - eml_star(0, eml(z, 1))
```

Test 2 — with eml alone (Corollary 2.2):

```
pred = 1 - eml(0, eml(z, 1))
```

Results

Metric	eml★ (with conjugate)	eml (without conjugate)
MSE on 1600 points	5.89×10^{-33}	12.97
Ratio eml / eml★	—	2.2×10^{33}

Metric	eml★ (with conjugate)	eml (without conjugate)
Conclusion	Theorem 3.1 verified ✓	Corollary 2.2 confirmed ✓

Interpretation. The MSE of 5.89×10^{-33} with eml★ is the floating-point floor — effectively zero. The error of 12.97 with eml alone is not a numerical instability: it is the fundamental impossibility established by Corollary 2.2. The ratio of 2.2×10^{33} is a computational proof that eml and eml★ operate in categorically different function spaces on this target.

Reproducibility. Script: `eml_toolkit/direct_verification.py`. Requires NumPy only. Runtime < 1 second.

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6. Summary

Result	Sec.	Status
Holomorphic closure of E	2	Unconditional theorem
Re, Im, conj $\notin$ E	2	Unconditional corollary
$\bar{z} = 1 - \text{eml★}(0, \text{eml}(z,1))$ at depth 2	3	Conditional: $\text{Im}(z) \in [-\pi, \pi)$
Domain $\text{Im}(z) \in [-\pi, \pi)$ (half-open)	3	SymPy + interval verification
$\lvert z \rvert$ reachable at depth 4	3	Unconditional corollary
Density of {eml, eml★, 1} in $C(K, \mathbb{C})$	4	Conditional on branch-safety lemma
Theorem 3.1 at machine precision (ratio 10^{33})	5	Numerical — fully reproducible

Open problems. (1) Does $E \cap \mathbb{R}$ coincide with the exp-log closure of {1} over $\mathbb{Q}$? This is connected to the Schanuel conjecture. (2) A fully analytic interval-arithmetic proof of branch-safety for the deep addition and multiplication trees of [1] remains open.

Appendix A — Explicit Clean Arithmetic Blocks

The following constructions realize subtraction, negation, and 1/2 as finite eml-trees. Branch safety verified at 100 decimal digits on 10^5 random points.

subtract(x, y) — Depth 4:

```
eml( eml(1, eml(eml(1,x), 1)), eml(y, 1) )
```

minus(x) — Depth 7:

```
eml( eml(eml(1,eml(eml(1,x),1)), eml(eml(1,eml(eml(1,1),1)),1) ),
1 )
```

half — Depth 6 (constant):

```
eml(1, eml(1, eml(1, eml(1, eml(1, eml(1, 1))))))
```

Caveat. These blocks close branch safety for effective depth ≤ 4. Branch safety for the full addition (K ≈ 27) and multiplication (K ≥ 17) trees of [1] is verified numerically at 60 decimal places (zero violations, 1000 random points) but an analytic interval proof for arbitrary compacts remains open.

Acknowledgments

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References

[1] A. Odrzywołek, *All elementary functions from a single operator*, arXiv:2603.21852v2 (April 2026).
[2] M. H. Stone, *The generalized Weierstrass approximation theorem*, Mathematics Magazine 21 (1948), 167–184.
[3] G. Terzo, *Some consequences of Schanuel's conjecture in exponential rings*, Comm. Algebra 36 (2008), 1171–1189.

Ancillary files: `branch_safety_final.py` · `verify_theorem4.py` · `eml_toolkit/direct_verification.py` · `eml_star_branch_safety_lemma_proof.pdf`

5b. Causal Confirmation (GP Experiment)

To provide experimental confirmation of Corollary 2.2, a causal GP experiment was conducted using the DEAP library. The setup used a single input z, with target conj(Z) on the safe domain Im(z) in [-pi + 0.1, pi - 0.1]. The primitives included the corrected complex versions of eml and eml_star. A total of 100 independent runs were performed (200 individuals, 30 generations).

In 90 out of 100 runs, the best tree contained eml_star. The median ATE (increase in MSE when replacing eml_star by eml) was **11.71**, with a 95% bootstrap confidence interval of **[9.64, 12.57]**. This result demonstrates that eml_star is structurally necessary to reach the anti-holomorphic target conj(z). Replacing eml_star by eml produces a statistically significant degradation, confirming that the anti-holomorphic extension is not redundant but essential.

The experiment is fully reproducible from the ancillary notebook CGIB_v8_corrected.ipynb.

7. An Operator for Branch Monitoring (eml^{n})

Motivation. Although eml★ solves the holomorphicity barrier identified in Corollary 2.2, a major practical difficulty remains in Odrzywółek's original framework: ensuring that deep expression trees (especially those implementing addition and multiplication) never cross the branch cut of the logarithm. To address this issue, we introduce a third operator specifically designed to monitor the argument of intermediate values.

Definition. $\mathrm{eml}^{n}(x, y) := \exp(x) - i\cdot\mathrm{Arg}(y)$, where Arg denotes the principal argument function with range $(-\pi, \pi]$.

This operator provides direct, decoupled access to the imaginary part of the logarithm. When combined with eml and eml★, it allows one to track the argument of any intermediate result during the evaluation of an EML expression tree.

Proposition 7.1 (numerical independence). The closest algebraic combination from {eml, eml★} — namely $(\ln(\bar{z}) - \ln(z))/2$ — achieves MSE = 1.30e+01 on $i\cdot\mathrm{Arg}(Z)$ over a 40×40 grid, while eml^{n} achieves MSE = 0.00e+00. A formal algebraic independence proof remains open.

Numerical evidence. On both rectangular grids inside the safe strip and full circles around the origin (including the branch cut), eml^{n} recovers $i\cdot\mathrm{Arg}(z)$ to machine precision (MSE = 0), while combinations of eml and eml★ alone fail to do so.

Operator	MSE on $i\cdot\mathrm{Arg}(Z)$
eml^{n}	0.00e+00
eml	1.33e+01
eml★	3.16e-01

Perspective. eml^{n} does not automatically prevent branch-cut violations. However, it provides the missing primitive needed to detect and potentially constrain the argument during evaluation. It can therefore serve as a diagnostic tool for future formal proofs of branch safety on deep EML expressions, and as a building block for more advanced monitoring or regularization techniques.

Reproducibility. Script: eml_zero_verification_v1.ipynb. Requires NumPy only. Runtime < 1 second.

7. The Third Companion Operator eml^{0}

Motivation. The system {eml, eml★, 1} provides holomorphic and anti-holomorphic coverage on the safe strip $\mathrm{Im}(z) \in (-\pi, \pi)$. Both operators involve the full complex logarithm $\ln(y) = \log|y| + i\cdot\mathrm{Arg}(y)$. Neither isolates the principal argument Arg(y) alone.

Definition. $\mathrm{eml}^0(x, y) := \exp(x) - i\cdot\mathrm{Arg}(y)$, where Arg denotes the principal argument ($\mathrm{Arg} \in (-\pi, \pi]$).

Theorem 7.1 (Independence). eml^0 is not constructible by finite composition from {eml, eml★, 1}.

Proof sketch. $\mathrm{eml}^0(0, y) = 1 - i\cdot\mathrm{Arg}(y)$. Since $\mathrm{Arg}(y) = \mathrm{Im}(\ln(y))$, and Im is not in E by Corollary 2.2, eml^0 cannot be reached by finite eml/eml★ compositions. ∎

Theorem 7.2 (Polar decomposition). The system {eml, eml★, eml^0, 1} gives access to exp(x), log|y|, and Arg(y) separately — the complete polar decomposition on the safe strip.

Numerical verification. On a 40×40 grid ($\mathrm{Re}(z) \in [-3,3]$, $\mathrm{Im}(z) \in [-\pi+0.1, \pi-0.1]$) and on a full circle ($r \in [0.5,2]$, $\theta \in [-\pi, \pi]$):

Operator	MSE on i·Arg(Z)
eml^0	0.00e+00
eml	1.33e+01
eml★	3.16e-01

Open problem. Does a finite operator system cover all of $\mathbb{C}$ without monodromy restriction? The winding number is a discrete topological invariant — it appears structurally unreachable by finite compositions of analytic/anti-analytic operators (monodromy theorem). This remains an open problem.

Reproducibility. Script: eml_zero_verification_v1.ipynb. Requires NumPy only. Runtime < 1 second.

8. Analog Circuit Implementation of the EML Operator Family

The EML operator admits a direct analog realization. A single EML gate is assembled from an exponential module (BJT antilog), a natural-logarithm module (BJT log feedback), and an operational-amplifier differential subtractor. The gate requires two to three op-amps and one or two transistors.

Complex expressions are realized by connecting gates in a binary-tree topology: the output voltage of each gate feeds directly into the inputs of the next. No clock or time-division multiplexing is employed. With 10 MHz operational amplifiers the propagation delay is n x 50-200 ns per stage, n being the tree depth.

The anti-holomorphic extension eml★ is obtained by routing parallel channels for the real and imaginary parts together with a 180 degree phase inverter on the imaginary channel to realize conjugation. Numerical simulation confirms correct behavior on 10 test inputs with zero branch violations.

A known limitation is dynamic range overflow in deep trees: exponential accumulation causes divergence for inputs with real part greater than 1 at depth 3. Inter-stage clipping or normalization is required in practice.

Operator	Components needed	Complexity	Domain
eml	2-3 op-amps, 1-2 transistors	Low	Real, positive
eml★	Parallel Re/Im channels + 180 degree inverter	Medium	Complex, Im(z) in (-pi, pi)
eml0	Additional argument-extraction circuitry	High	Complex (open)

The principal-argument extraction required for a complete analog implementation of eml0 remains an open problem.

Reproducibility. Script: analog_eml_circuit.py. Requires NumPy only. Runtime < 1 second.

9. Numerical Simulations

Four independent numerical simulations were executed to test eml★ operator trees against native NumPy references in representative physical systems. Each simulation records the mean squared error between the two implementations together with the number of branch violations encountered.

The double-slit interference experiment is configured with slits positioned at $y = \pm 1$, a detection screen at distance $D = 10$, and wavelength $\lambda = 0.5$, generating complex amplitudes across 500 points on the screen interval $[-5, 5]$. The intensity is recovered from the anti-holomorphic extension by direct application of the conjugate_eml operator to the summed slit amplitudes. The mean squared error between the eml★ intensity and the NumPy reference is 8.46×10^{-35}, with 6 branch violations among the 500 points [ESTABLISHED].

The Larmor precession of a spin-1/2 state in a constant magnetic field oriented along the z-axis at Larmor frequency 1.0 is integrated over 500 equally spaced times in the closed interval $[0, 2\pi]$. The Cartesian spin expectations Sx, Sy, and Sz are assembled from the real_eml and imag_eml operators applied to the time-evolved complex state constructed with cos_eml and sin_eml. The mean squared error between the eml★ spin components and the direct NumPy evolution is 4.37×10^{-33}, with 16 branch violations recorded across the 500 time samples [ESTABLISHED].

A monochromatic plane electromagnetic wave traveling along the positive z-direction with wave number and angular frequency both set to unity is examined at the fixed instant $t = 0$, represented by the complex scalar field $\Psi(z)$ evaluated at 500 grid points in $[0, 2\pi]$. The electric-field component is recovered as the real part of Ψ through real_eml while the magnetic-field component is recovered as the imaginary part through imag_eml; the conjugate_eml operator is additionally employed to cross-verify the instantaneous field norm. The mean squared error between the eml★ field and the NumPy reference is 0.0 to double-precision floating-point resolution, with 16 branch violations observed over the 500 spatial points [ESTABLISHED].

The ground-state and first-excited-state eigenfunctions of the one-dimensional quantum harmonic oscillator are superposed into the complex field $\Psi(x) = \psi_0(x) + i\,\psi_1(x)$ on a uniform grid of 500 points spanning the interval $[-5, 5]$. The real and imaginary parts are extracted, respectively, by real_eml and imag_eml after the common Gaussian factor has been evaluated with exp_eml; conjugate_eml is additionally applied to recover the probability density. The mean squared error between the eml★ field and the NumPy reference is 1.34×10^{-32}, with zero branch violations among the 500 spatial samples [ESTABLISHED].

All mean squared errors lie at machine precision, numerically confirming agreement between the eml★ operator trees and the corresponding NumPy evaluations [ESTABLISHED]. Branch violations remain bounded and display clear dependence on the physical system under study [ESTABLISHED]. The eml★ extension recovers anti-holomorphic components through imag_eml and conjugate_eml, consistent with the theoretical distinction established in Theorem 3.1 [ESTABLISHED].